

Case Study: I4C Cyber Fraud Investigation and Data Security

Case Study Title

Securing Cybercrime Investigation Data Using Cryptographic Techniques

Background

The **Indian Cyber Crime Coordination Centre (I4C)** works as a national-level framework for dealing with cybercrime-related challenges and supporting coordination among agencies. In a simulated cybercrime investigation, sensitive information such as complaint details, victim information, transaction details, suspect information, and investigation records must be protected from unauthorized access.

You are working as a **Junior Cybersecurity Analyst** in a simulated I4C cybercrime investigation team.

The investigation team has collected the following fictional information:

Complaint ID: I4C-2026-001

Victim Name: Rahul Patel

Fraud Amount: ₹50,000

Suspect ID: SUS-1025

Investigation Status: Active

Officer ID: OFF-205

The team wants to develop a small security prototype to protect this information.

Definitions You Need to Understand

1. Encryption

Definition: Encryption is the process of converting readable data (plaintext) into an unreadable form (ciphertext) using an algorithm and a key.

Student Task: Find out how encryption protects sensitive cybercrime investigation information.

2. Classical Encryption

Definition: Classical encryption refers to traditional cryptographic techniques that transform plaintext into ciphertext using simple mathematical or substitution methods.

Examples:

- Caesar Cipher
- Substitution Cipher

Student Task: Research the working principle, advantages, limitations, and security weaknesses of classical encryption.

3. Symmetric Encryption

Definition: Symmetric encryption uses the **same secret key** for encryption and decryption.

Student Task: Explain why symmetric encryption is suitable for protecting large amounts of investigation data.

4. AES

Definition: AES (Advanced Encryption Standard) is a modern symmetric-key encryption algorithm used to protect digital information.

Student Task: Investigate how AES works and implement encryption and decryption of the fictional investigation record.

5. Asymmetric Encryption

Definition: Asymmetric encryption uses a pair of keys:

- Public Key
- Private Key

Student Task: Explain how public-key and private-key cryptography solves the problem of securely sharing information or keys.

6. RSA

Definition: RSA is an asymmetric cryptographic algorithm that uses public and private keys for cryptographic operations.

Student Task: Implement RSA and investigate how it can be used to protect the AES key in the case study.

7. Hashing

Definition: Hashing converts input data into a fixed-length hash value using a hash function.

Student Task: Explain why hashing is different from encryption and why a hash is normally not used to recover the original data.

8. Password Security

Definition: Password security involves protecting authentication credentials against unauthorized access and attacks.

Student Task: Investigate how password hashing can be used instead of storing passwords as plaintext.

Student Investigation Activities

Students must complete the following activities **independently**.

Activity 1 – Cybersecurity Environment Setup

Set up a basic cybersecurity programming environment.

Investigate

- Python installation
- VS Code/Jupyter Notebook
- **cryptography** library
- Python **hashlib**
- OpenSSL or another security tool

Submit

- Environment setup steps
- Tool names and versions
- Installation screenshots
- Successful execution screenshot

Activity 2 – Classical Encryption

Using the fictional investigation message:

CYBER CRIME INVESTIGATION

Tasks

1. Research Caesar Cipher.
2. Select an appropriate shift value.
3. Implement encryption.
4. Implement decryption.
5. Verify the original message.
6. Implement a substitution cipher.
7. Compare Caesar Cipher and Substitution Cipher.
8. Explain why these techniques are unsuitable for modern sensitive information.

Student Deliverable

Prepare a table:

Parameter	Caesar Cipher	Substitution Cipher
Type		
Key		
Encryption method		

Decryption method		
Security level		
Major limitation		

Activity 3 – AES Investigation Data Protection

The investigation team wants to protect:

Complaint ID

Victim Name

Fraud Amount

Suspect ID

Investigation Status

Tasks

1. Research AES.
2. Generate an AES key.
3. Encrypt the investigation data.
4. Display/store the ciphertext.
5. Decrypt the ciphertext.
6. Verify that the original data is recovered.
7. Explain why AES is more appropriate than Caesar Cipher.

Think About

What would happen if the AES secret key were stolen?

Activity 4 – RSA Secure Key Transmission

The investigation team cannot simply send the AES secret key openly to another authorized officer.

Tasks

1. Research RSA.
2. Generate an RSA public/private key pair.
3. Understand the role of each key.
4. Use RSA to protect the AES key.
5. Recover the AES key using the appropriate RSA key.
6. Use the recovered AES key to decrypt the investigation data.

Research Question

Why is a hybrid approach using AES + RSA more practical than using RSA to encrypt the entire investigation file?

Activity 5 – Hashing and Password Security

The investigation portal requires an officer to authenticate before accessing the case record.

Use a **fictional password** for experimentation.

Tasks

1. Research MD5.
2. Generate an MD5 hash.
3. Research SHA-256.
4. Generate a SHA-256 hash.
5. Implement password verification.
6. Compare MD5 and SHA-256.
7. Investigate why raw MD5 and SHA-256 are not recommended for modern password storage.
8. Research **Argon2, bcrypt, scrypt, and PBKDF2** as modern password-hashing approaches.

Important Rule

Do not use your personal password, real credentials, real financial information, or real victim information.

Final Integrated Challenge

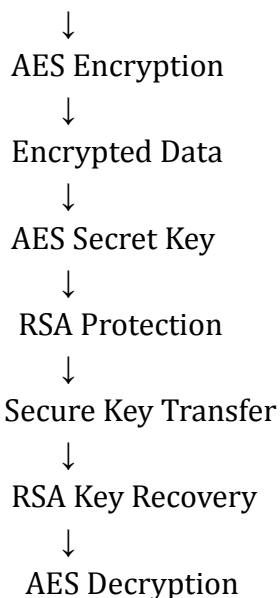
Now combine all five activities into **one cybersecurity workflow**.

Scenario

An authorized I4C investigation officer needs to access a protected cybercrime case.

Your system should conceptually demonstrate:

Cybercrime Investigation Data



↓
Original Case Data

Authentication should be handled separately through a secure password-verification mechanism.

Final Student Submission

Students should submit **one PDF report** containing:

1. Case study understanding
2. Definitions
3. Environment setup
4. Classical encryption implementation
5. AES implementation
6. RSA implementation
7. Hashing implementation
8. Comparison of cryptographic techniques
9. Integrated security workflow/diagram
10. Screenshots of execution
11. Answers to self-study questions
12. Security observations
13. Conclusion
14. References